



Image-Based Defect Detection System for Manufacturing Inspection

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ImageDetect5

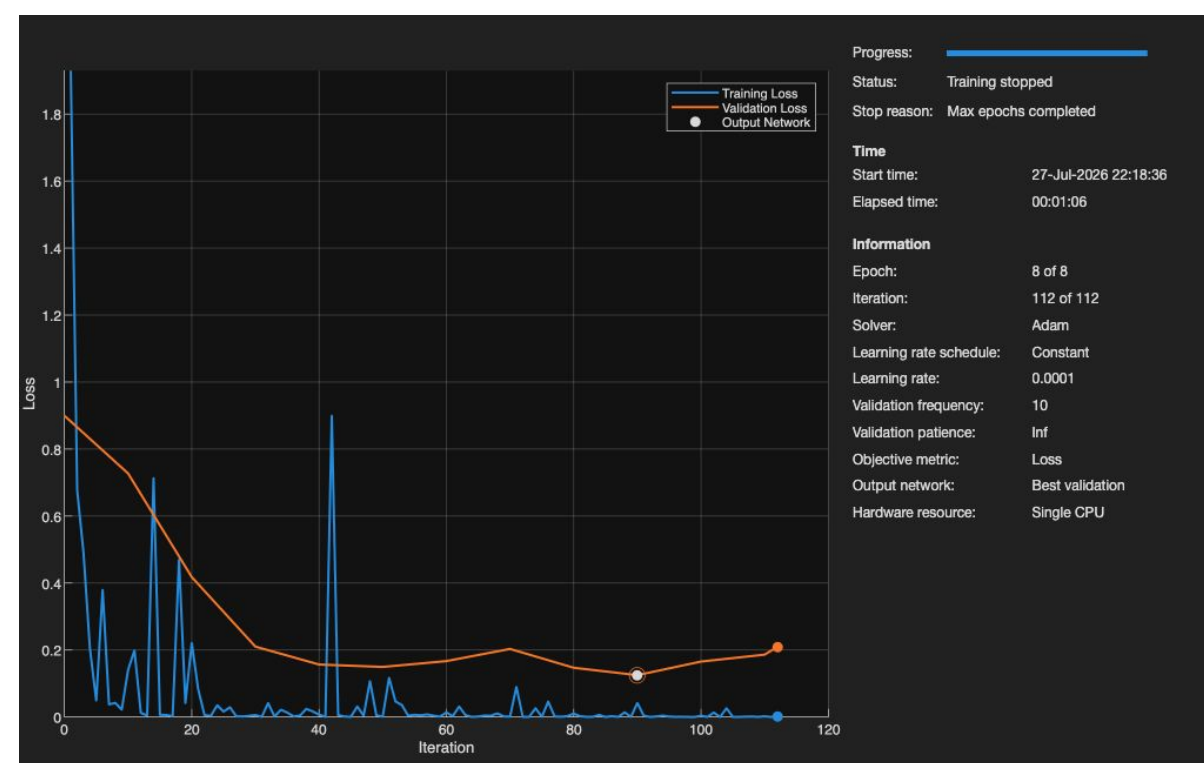


Introduction

Image-based defect detection uses computer vision and machine learning to automatically identify defects in manufactured products. This is important because automated inspection can improve product quality, reduce human error, lower costs, and make manufacturing more efficient. This project investigates whether a MATLAB-based image classification model can accurately detect defective products from images.

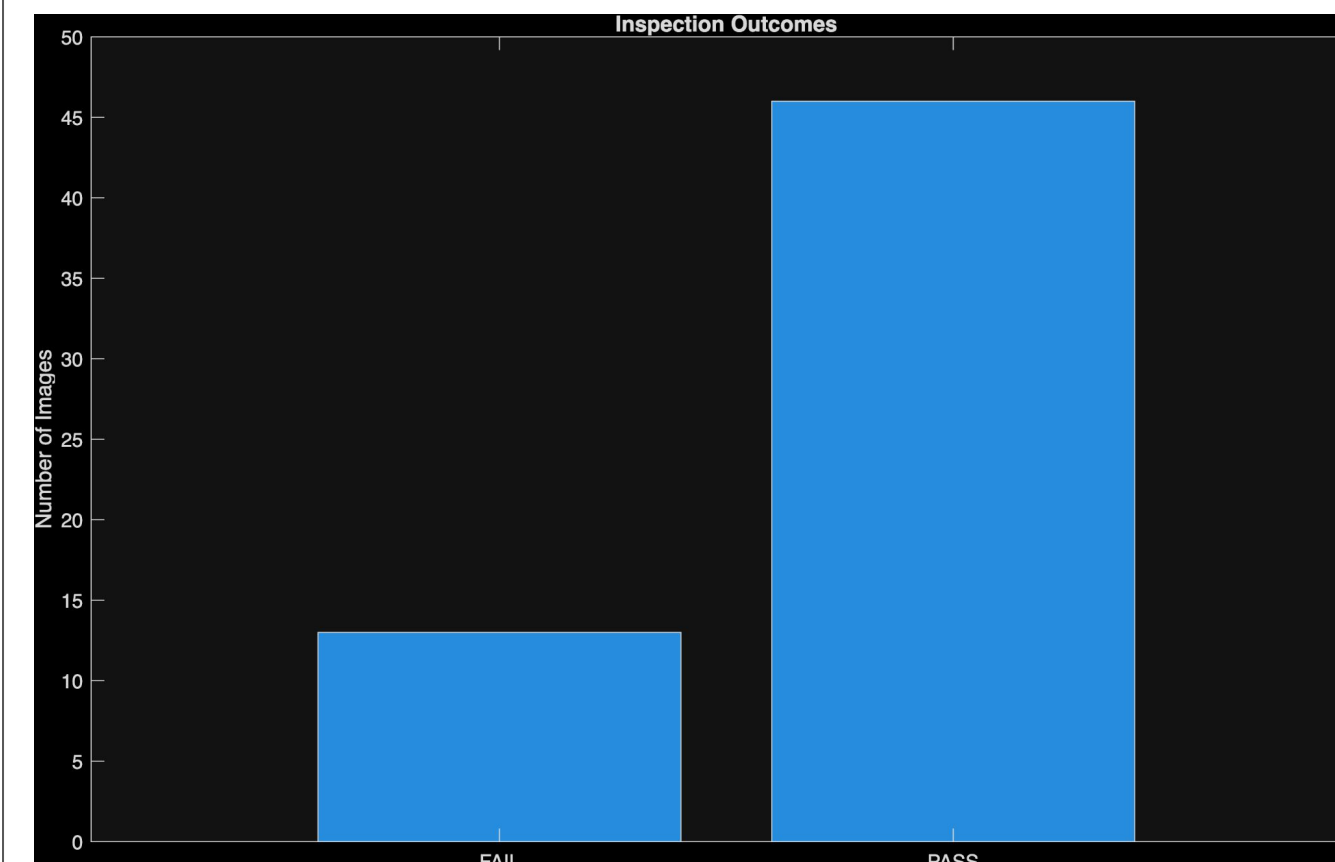


Methods



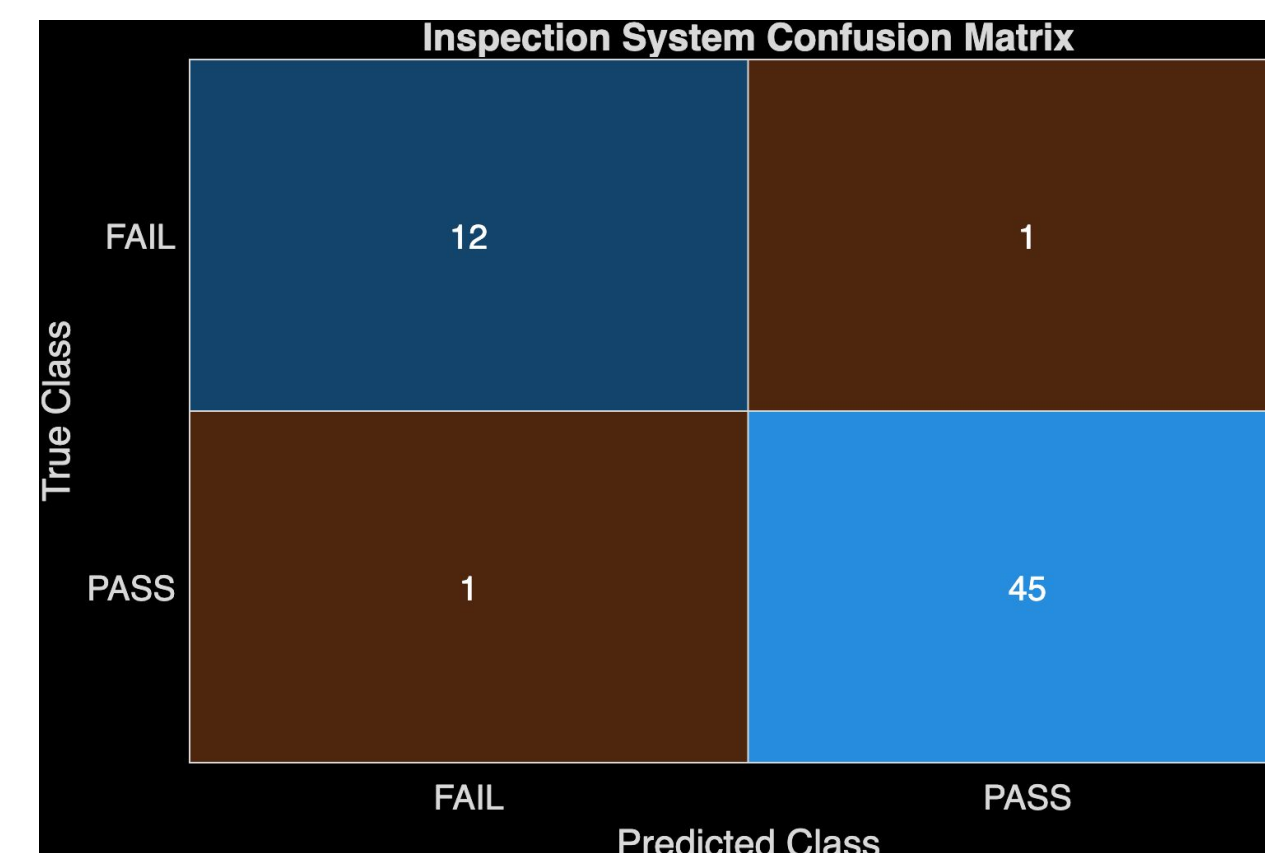
The project was developed in MATLAB using the Deep Learning Toolbox Model for the ResNet-18 network to classify images of defective and non-defective products. A baseline rule-based system was also implemented as a failsafe to verify the AI model's predictions, improving the reliability, robustness, and accuracy of the manufacturing inspection process.

Results



The inspection results showed that a total of 59 images were processed. Of these, 46 images passed the inspection, while 13 images failed.

The inspection system's confusion matrix shows that, of the 13 images classified as failed, 12 were correctly identified as true failures, while 1 was actually a true pass (false positive). Of the 46 images classified as passed, 45 were correctly identified as true passes, while 1 was actually a true failure (false negative).



Under simulated brightness, blur, and noise, the system achieved 96.6% accuracy under baseline and brightness conditions and 98.3% under blur. Noise reduced accuracy to 89.8% and raised the false-reject rate to 13.0%, but all defects were still detected. Overall, the system was robust to brightness and blur but more sensitive to noise.

Discussion

The results suggest that AI can effectively detect manufacturing defects and support automated quality inspection. While the model performed well, its accuracy is limited by the size and diversity of the training data. Future work would involve training the model with a larger and more diverse dataset and testing it on a much larger number of production images to better evaluate its accuracy, reliability, and scalability in a real manufacturing environment.

References

Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger, "A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection", IEEE Conference on Computer Vision and Pattern Recognition, 2019

Acknowledgements

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